

Ulysse Hennebelle

Ph.D. Student in Operations Research

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EDUCATION

Cornell University - Cornell Tech

Ph.D. Student in Operations Research and Information Engineering

2024-Present

New York, NY

Advisor: Omar El Housni.

Selected coursework: Analysis of Algorithms, Convex Analysis, Integer Programming, Online Algorithms and Market Design, Probability II, Randomized Algorithms.

École Polytechnique

Ingénieur Polytechnicien (Bachelor's and Master's degree)

2020-2024

Palaiseau, France

Final rank: 34/435.

Selected coursework: Operations Research, Statistical Modeling of Processes, Algebraic Topology, Quantum Physics, Competitive Programming, Deep Learning.

RESEARCH INTERESTS

Assortment Optimization, Choice Modeling, Revenue Management, Online and Stochastic Decision-Making, Data-driven Optimization, Optimization under Uncertainty.

PAPERS

Published

[1] Two-sided Assortment Optimization: Adaptivity Gaps and Approximation Algorithms

with Omar El Housni and Alfredo Torrico.

Operations Research.

Preliminary version: Adaptivity Gaps in Two-Sided Assortment Optimization, IPCO 2024, pp. 139-153.

[2] Runtime Analyses of Multi-Objective Evolutionary Algorithms in the Presence of Noise

with Matthieu Dinot, Benjamin Doerr, and Sebastian Will.

Proceedings of IJCAI 2023, pp. 5549-5557.

A short version appeared in the GECCO Companion 2024, pp. 33-34.

Working Papers

[3] Condorcet-Winning Sets and Peer Selection in Planar Metric Elections

with Gabriel de Azevedo.

arXiv preprint arXiv:2608.27653, 2026.

[4] A Unified Framework for Assortment Optimization and Matching under MNL Choice

Work in progress.

TEACHING EXPERIENCE

Teaching assistant for graduate (M.Eng.) courses at Cornell Tech, serving classes of 60+ students.

ORIE 5530 Modeling Under Uncertainty

Fall 2026

Teaching Assistant, Graduate (M.Eng.)

CS 5854/ORIE 5138 Networks and Markets

Spring 2026

Teaching Assistant, Graduate (M.Eng.)

ORIE 5132 Pricing Analytics and Revenue Management

Spring 2026

Teaching Assistant, Graduate (M.Eng.)

ORIE 5530 Modeling Under Uncertainty

Fall 2025

Teaching Assistant, Graduate (M.Eng.)

ORIE 5600 Financial Engineering with Stochastic Calculus

Fall 2024

Teaching Assistant, Graduate (M.Eng.)

INDUSTRY & RESEARCH EXPERIENCE

Data Scientist Intern

May-August 2026

New York City Police Department | Siegel PiTech PhD Impact Fellow

New York, NY

- Built a Python pipeline for four years of 6.0M NYC 311 and 17.0M NYPD 911 calls; assigned calls to 303 patrol sectors and produced a 1.33M-row sector-by-shift panel.

- Engineered rolling forecasts at 1-, 14-, and 28-day horizons using 117 history, calendar, location, and weather inputs; grouped SHAP attributions into four interpretable feature families.
- Benchmarked Histogram Gradient Boosting, Prophet, and Ridge across three shifts, three horizons, and three objectives. In the sampled presentation benchmark, Gradient Boosting outperformed Prophet in all nine target-horizon comparisons (mean global R-squared 0.53 vs. 0.50).
- Assigned calls to census blocks within police precincts and implemented Kulldorff's scan statistic to identify chronic locations with unusually high call volumes.
- Extended the scan statistic from circular regions to arbitrary convex shapes. Convex regions achieved a higher likelihood-ratio fit in 93.4% of 1,404 paired weekly comparisons and tied in 6.6%; implementation improvements increased speed by approximately 100x.

Research Assistant

Cornell Tech

Mar-Jul 2023

New York, NY

- Early research on two-sided assortment optimization; led to an Operations Research paper and a preliminary conference version at IPCO 2024.

Visiting Associate

Boston Consulting Group

Jun-Aug 2022

Casablanca, Morocco

- Worked with BCG Digital Ventures on strategy and venture-building projects for large clients.

HONORS AND AWARDS

Siegel PiTech PhD Impact Fellowship

Summer 2026

Professor Robert E. Bechhofer Fellowship, Cornell University

Spring 2025

TALKS

- INFORMS Annual Meeting, San Francisco, CA, 2026. **Online Matching with MNL Choices.**
- INFORMS Annual Meeting, Atlanta, GA, 2025. **Constrained Two-Sided Assortment Optimization.**
- Revenue Management and Pricing Conference, New York, NY, 2025. **Constrained Two-Sided Assortment Optimization.**

LEADERSHIP AND SERVICE

Voting Student Representative

Institut Polytechnique de Paris Board of Directors

2021-2023

Palaiseau, France

- Represented an 8,000-student body on the board of directors.

Officer

French Army

Sep 2020-Apr 2021

France

- Led a platoon of 28 recruits and managed military training, discipline, safety, and well-being.

TECHNICAL SKILLS AND LANGUAGES

Languages: French (native); English (fluent).

Programming and data: Python, NumPy, pandas, scikit-learn, C++, Java, R, OCaml.

Optimization: Gurobi, CVXPY.

Tools: Git, LaTeX, Codex.

Interests: GeoGuessr (online geography game).